PropertyScope NSW

Release 2 / Technical report

41026 Advanced Software Development — Assessment 3

Submission	Group 20
Team	Matthew Shelton, Burhan Naeem, James Huang, Michael White, Derek Song
Due	25 October 2026, 11:59 pm Sydney time
Repository	MattShelton04/41026ASDProject
Showcase video	Pending
Showcase	23 October 2026
Commit reference	main

TO COMPLETE / GROUP

Replace "Pending" with the published showcase video link, and use the same URL in RELEASE2_METADATA and Section 14

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1 Project overview

PropertyScope NSW helps buyers research NSW property. Each student owns a containerised frontend, backend/API and database slice behind a shared home page, and Matthew also leads the shared platform. Release 1 added host MCP and RAG servers to the Release 0 AI-mode and its Plan/Act/Observe/Adapt loop. Release 2 keeps all of these and adds:

- a shared host Multi-Agent Server (Planner → Worker → Reviewer → human review) that each feature calls through its own backend;
- three agentic-loop review modes: multi-agent workflow, testing report and cloud deployment;
- pre-commit security scans, and post-commit tests of two endpoint functions per student;
- an Azure deployment workflow with the AI tier disabled by default.

Student	Feature	Release 2 responsibility
1 Matthew Shelton (24763373)	F1 Data platform and property discovery	Shared platform: Multi-Agent Server, review modes, security scans, endpoint test kit, Azure. F1 readiness-review workflow and endpoint tests
2 Burhan Naeem (24764134)	F2 Property sales and market cases	F2 workflow, endpoint tests and evidence
3 James Huang (24970865)	F3 Suburb, crime and liveability analytics	F3 workflow, endpoint tests and evidence
4 Michael White (24846267)	F4 Site, planning and building due diligence	F4 workflow, endpoint tests and evidence
5 Derek Song (24833978)	F5 Buyer journey workspace	F5 workflow, endpoint tests and evidence

2 Final repository structure

Path	Contents	Release
`student-1/` … `student-5/`	Feature slices: UI, API, database service, tools, corpus; `config/multi-agent/` and `tests/endpoints/`	0–2
`ai-services/ai-mode/`, `agent-core/`	AI-mode run API and the shared agentic loop	0
`ai-services/mcp-server/`, `rag-server/`	Tool transport and local retrieval	1
`ai-services/multi-agent-server/`	Planner, Worker, Reviewer, human decisions, JSONL audit, CLI	2
`scripts/devtools/review/`	`ai review` modes and the release decision	2
`shared/frontend/multi-agent/`, `shared/testkit/`	Shared workflow panel; endpoint client, JUnit summary, `FakeMultiAgentServer`	2
`.pre-commit-config.yaml`, `scripts/security/`	Security hooks and report	2
`.github/workflows/`	`student-N.yml`, `integration-ci.yml`; `cloud-deployment.yml`	0–2
`deployment/azure/`, `docker-compose.azure.yml`	Bicep, VM scripts, TLS edge, systemd AI units	2
`docs/release-2/evidence/`	Security, CI, multi-agent, review, local and cloud evidence	2

3 Release 2 design flow

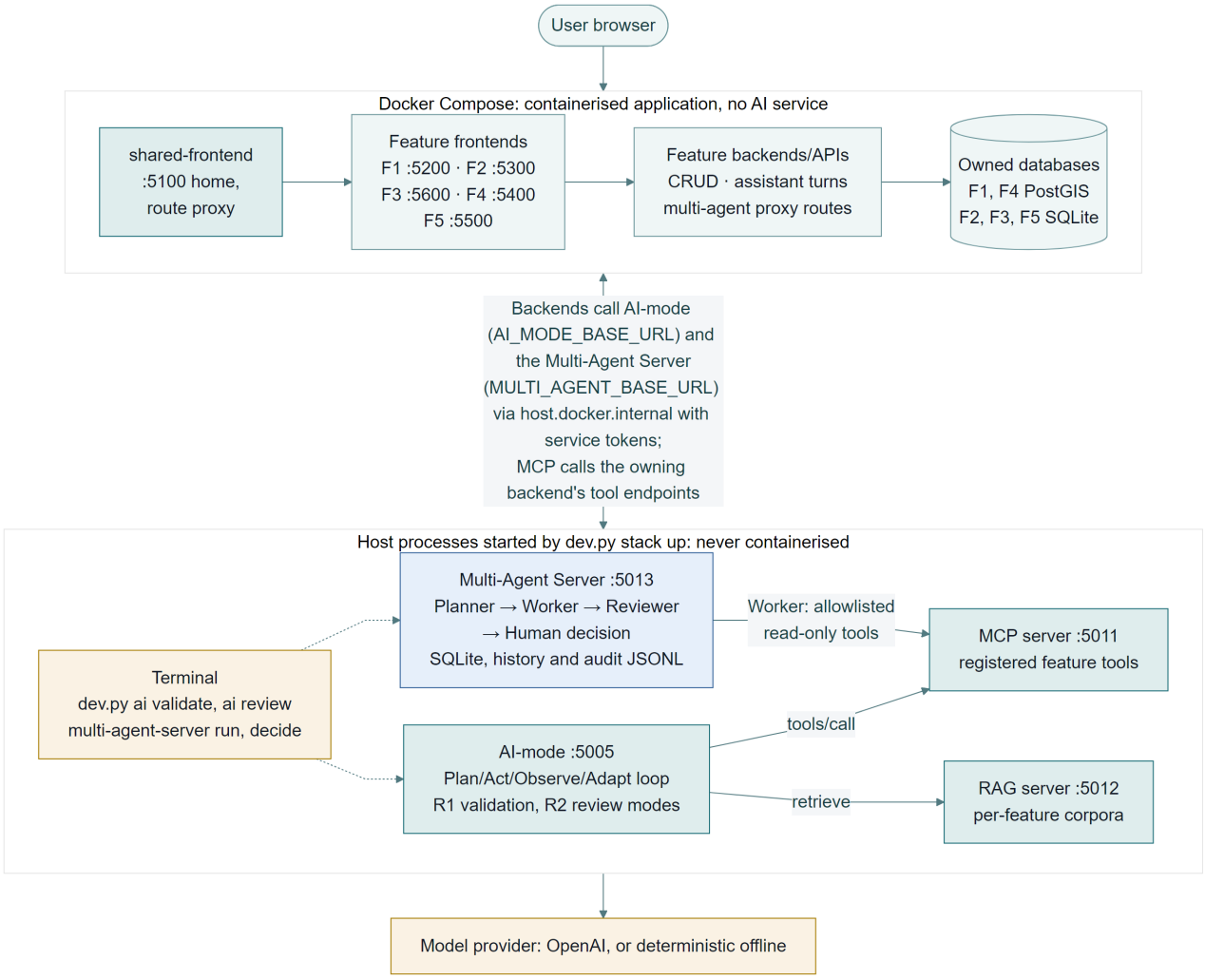


Figure 1 Release 2 integrated architecture

CRUD requests go from the browser through the shared edge to a feature frontend, its backend/API and the database service it owns. The Release 1 flows are unchanged: an assistant turn goes from the backend to AI-mode, which calls that feature's MCP tools and retrieves from its RAG corpus. In Release 2 each backend also proxies the Multi-Agent Server over HTTP with `MULTI_AGENT_BASE_URL` and a service token. The Planner plans read-only steps, the Worker gathers evidence through MCP, the Reviewer checks it, and only a human decision ends the run. The shared panel shows every stage. No AI service opens a feature database or runs in Docker.

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template
1	:5200 → `/api/data-platform/v1` → `f1-postgres`	14	19	`f1-release-readiness-review` (2 read-only tools)
2	:5300 → `/api/market-intelligence/v1` → `f2-db-api`	3	3	Not registered yet
3	:5600 → `/api/suburb-analytics/v1` → `f3-database`	4	8	Not registered yet

Student	Frontend → backend/API → database	MCP tools	RAG documents	Multi-agent template
4	:5400 → `/api/due-diligence/v1` → `f4-postgres`	3	3	Not registered yet
5	:5500 → `/api/buyer-workspaces/v1` → `f5-db-api`	5	5	Not registered yet

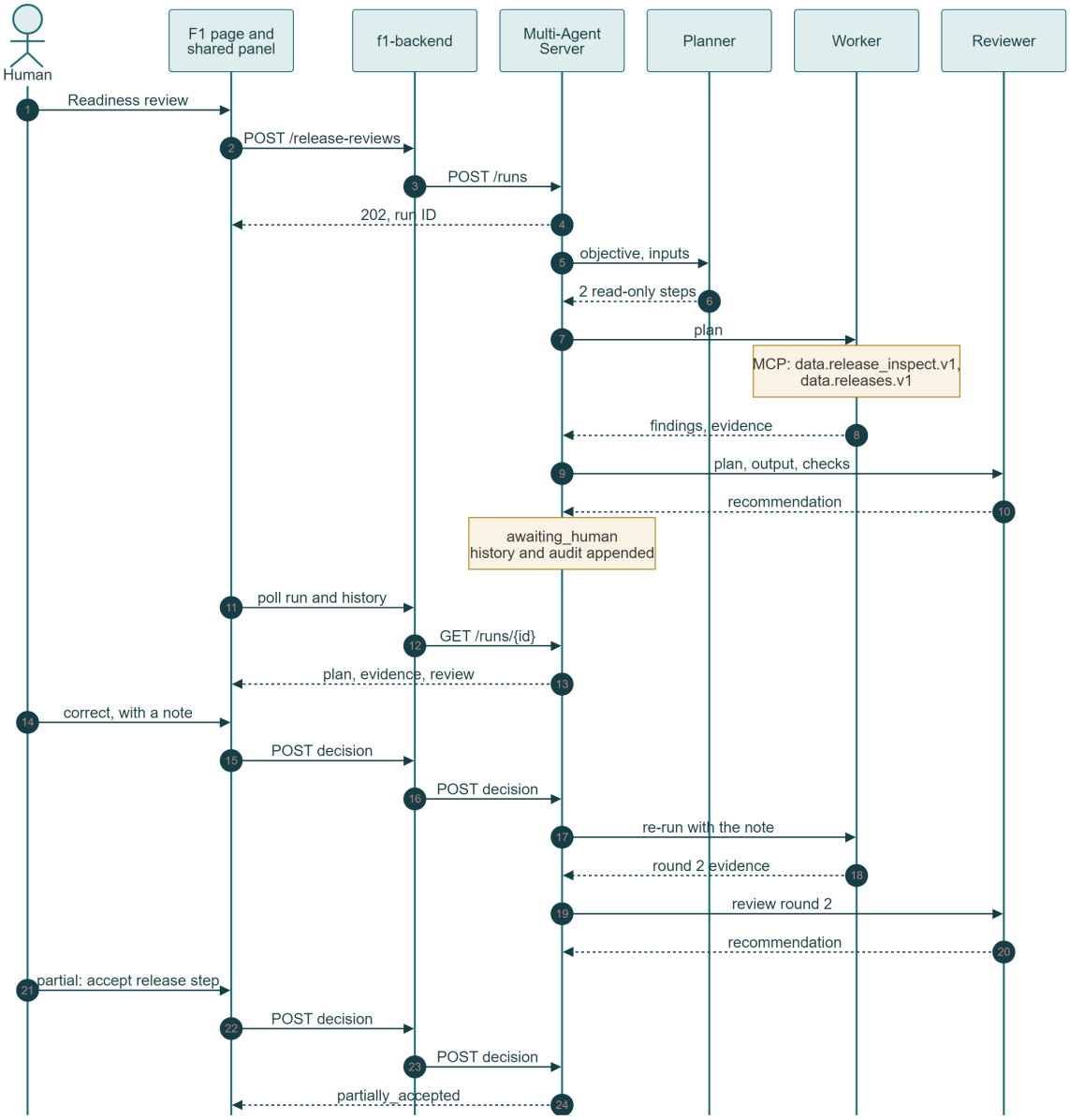


Figure 2 Multi-agent flow, using the Feature 1 readiness review

Student 1

The `f1-release-readiness-review` workflow recommends whether a candidate dataset release is ready to publish. Seven proxy routes under `/api/data-platform/v1/release-reviews` fix the template server-side and hide other features' runs. Its allowlist omits `data.release_publish.v1`, so a review never publishes (Figure 2).

Student 2

TO COMPLETE / BURHAN NAEEM

Describe F2's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 3

TO COMPLETE / JAMES HUANG

Describe F3's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 4

TO COMPLETE / MICHAEL WHITE

Describe F4's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

Student 5

TO COMPLETE / DEREK SONG

Describe F5's multi-agent design flow in about 40 words: template, read-only tools and proxy routes

4 Cloud deployment and CI/CD architecture

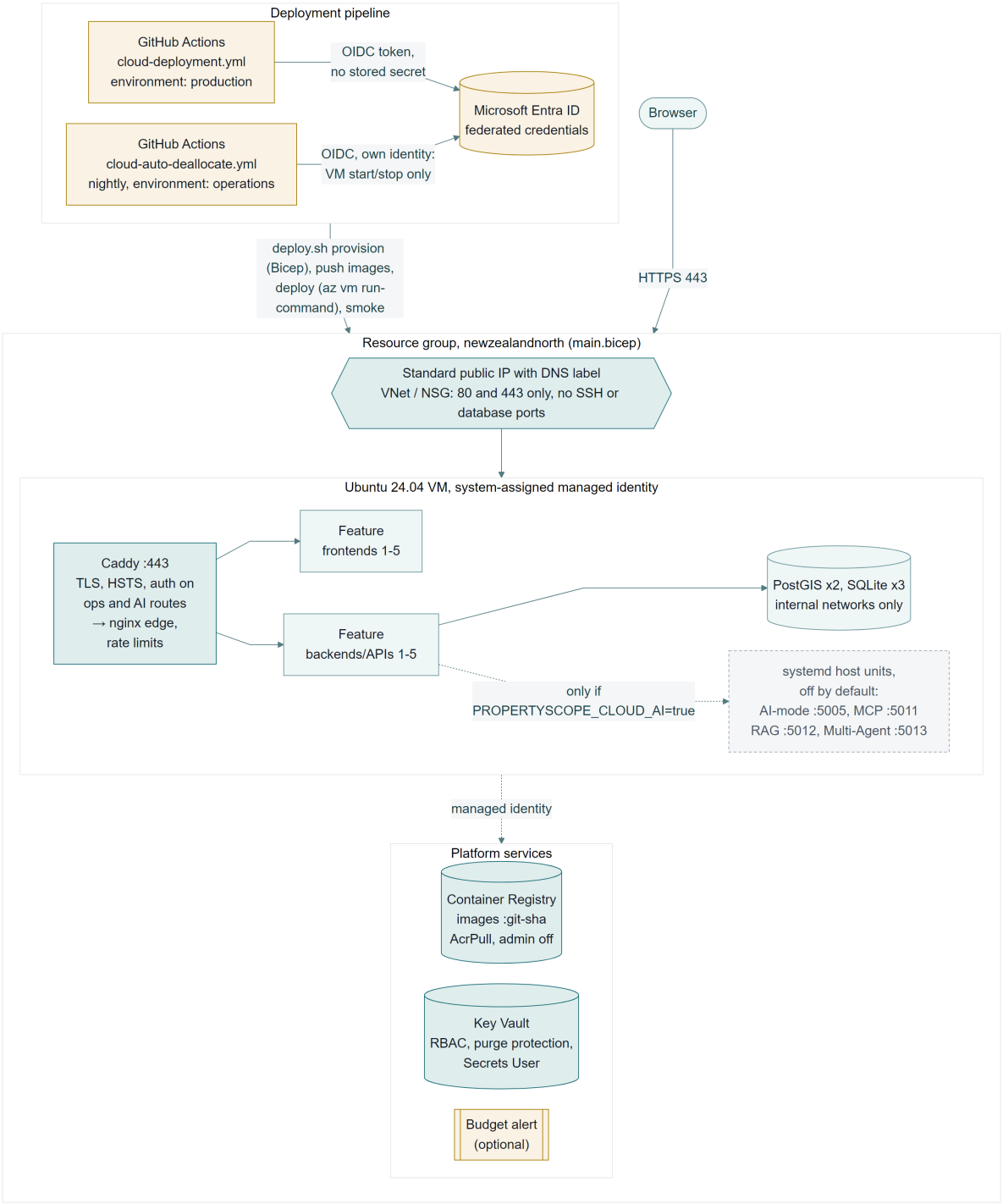


Figure 3 Azure reference architecture

We chose Azure (ADR-048). One Ubuntu VM runs the unchanged Compose model with a production override, because the AI tier must stay a host process and container-only services cannot host it.

- **Hosting:** images are built once per commit, pushed to Azure Container Registry tagged with the Git SHA and pulled by the VM's managed identity.
- **Endpoints:** a public IP with a DNS label; the NSG admits only 80 and 443. Caddy terminates TLS in front of the nginx edge; databases stay on internal networks.

- **Secrets:** Key Vault, read by the managed identity. GitHub signs in with OIDC.
- **AI tier:** AI-mode, MCP, RAG and Multi-Agent are systemd units, off unless `PROPERTYSCOPE_CLOUD_AI=true`.

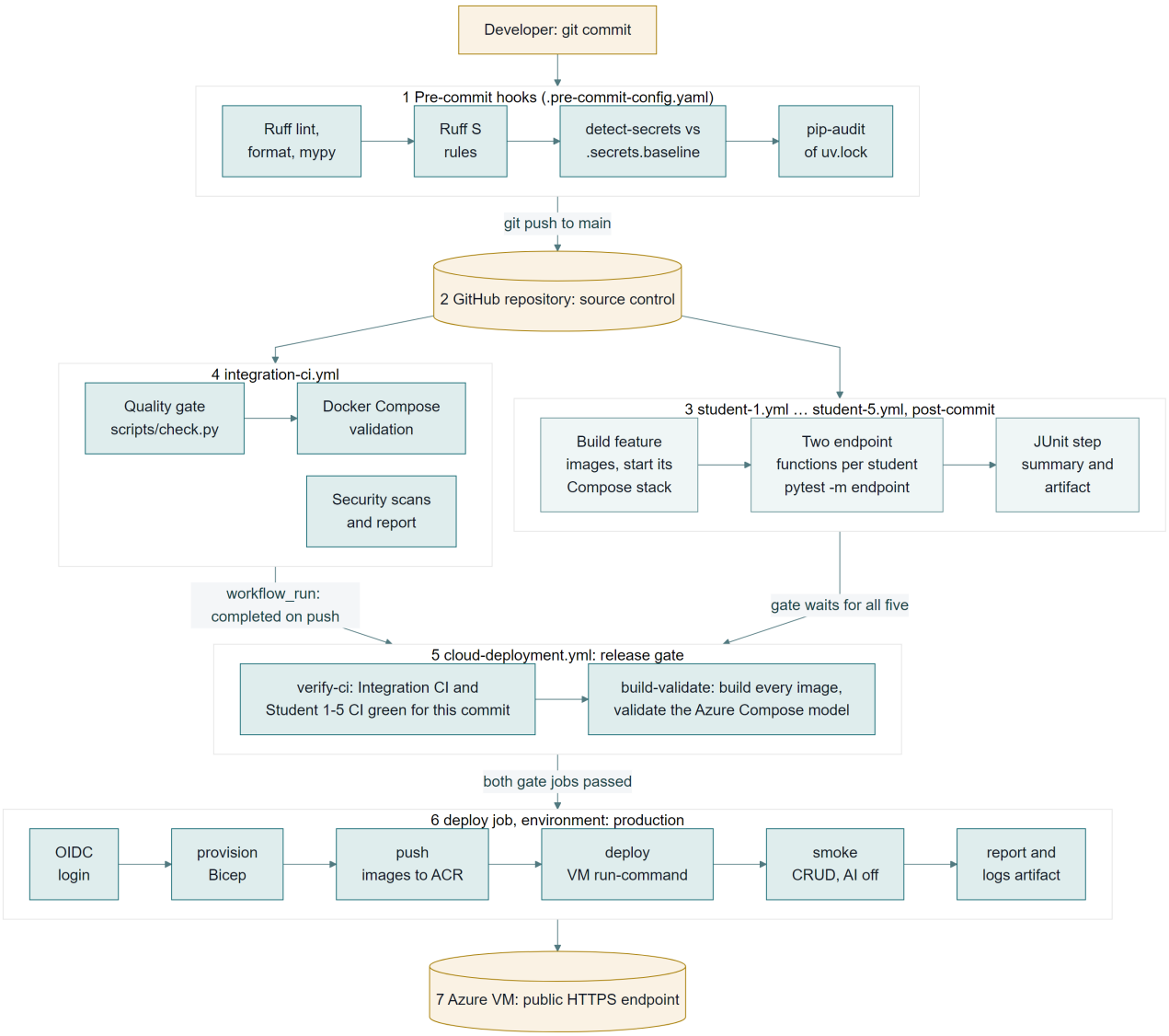


Figure 4 DevOps flow from commit to Azure

A commit runs the pre-commit hooks. A push to `main` runs `integration-ci.yml` and each `student-N.yml`, which tests two endpoint functions. `cloud-deployment.yml` then gates the release: `verify-ci` waits until Integration CI and all five student workflows have passed for that commit, and `build-validate` builds every image and validates the Azure Compose model. Only then does `deploy` run the reusable `deploy.sh` steps in the protected `production` environment.

The VM is a `Standard_D4as_v4` in `newzealandnorth`, because the *Azure for Students* region policy blocks `australiaeast`.

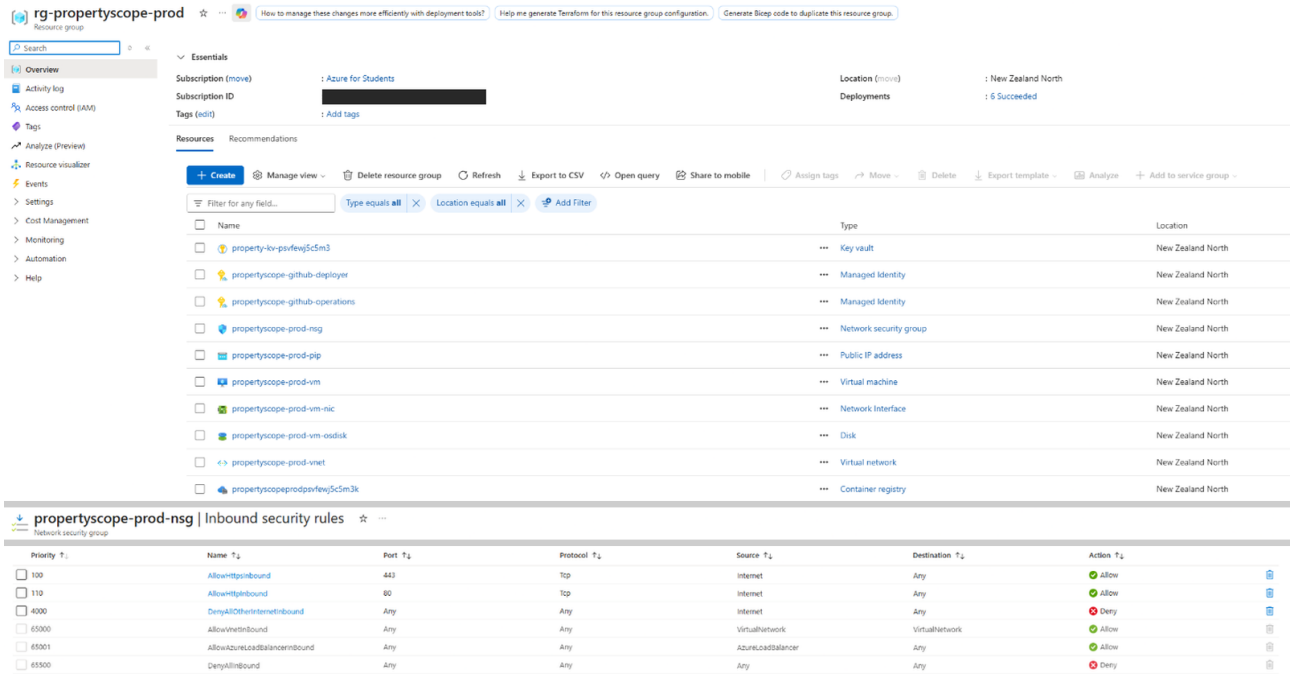


Figure 5 Deployed resources and inbound rules

5 Pre-commit security testing

`.pre-commit-config.yaml` runs three security hooks on every `git commit`, beside Ruff, formatting and mypy: Ruff S (flake8-bandit) rules, detect-secrets against the reviewed `.secrets.baseline`, and pip-audit of `uv.lock`. `integration-ci.yml` repeats them on every push. `dev.py security report` records results per slice and owner:

Security report: overall **PASS**, commit `1a68f0a`, 2026-10-16.

Scan	Result	Findings
Ruff	**PASS**	0 open; 27 fixed; 132 justified with `# noqa`; 85 pending owner
detect-secrets	**PASS**	0 new; 0 confirmed; 0 unaudited; 56 reviewed false positives in `.secrets.baseline`
pip-audit	**PASS**	126 locked packages audited; 0 open; 19 accepted risk(s) in 2 package(s)

Slice	Owner	Ruff open	Fixed	Noqa	Pending owner	Secrets to act on
Shared platform	Matthew Shelton	0	18	111	0	0
Student 1	Matthew Shelton	0	9	21	0	0
Student 2	Burhan Naeem	0	0	0	8	0
Student 3	James Huang	0	0	0	10	0
Student 4	Michael White	0	0	0	8	0
Student 5	Derek Song	0	0	0	59	0

In the recorded commit run, detect-secrets first blocked a commit. After seven false positives were reviewed into the baseline, the same commit passed:

```
security: Ruff flake8-bandit rules (S).....................Passed
security: detect-secrets against .secrets.baseline.............Passed
security: pip-audit of locked dependencies (uv.lock)............Passed
```

TO COMPLETE / MATTHEW SHELTON

Regenerate the security report on main at the evidence freeze; the committed one scanned a release-2/wrap-up-shared-f1 branch commit

TO COMPLETE / BURHAN NAEEM

Fix or justify F2's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / JAMES HUANG

Fix or justify F3's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / MICHAEL WHITE

Fix or justify F4's pending-owner Ruff S findings, then regenerate the report

TO COMPLETE / DEREK SONG

Fix or justify F5's pending-owner Ruff S findings, then regenerate the report

6 Post-commit CI/CD testing

Each `student-N.yml` starts its feature's Compose stack in GitHub Actions with the AI tier disabled and runs `pytest -m endpoint` against two endpoint functions of its backend/API. A JUnit step summary requires at least two passes and the XML is uploaded. `collect_ci_evidence.py` retains each run's record, reports and log under `evidence/ci/`.

Student	`student-N.yml` run	Conclusion	Commit	Endpoint tests
1	#353	success	`d449a43`	34 passed, 0 failed (2 reports) (record)
2	Pending	-	-	-
3	Pending	-	-	-
4	Pending	-	-	-
5	Pending	-	-	-

TO COMPLETE / BURHAN NAEEM

Record a green `student-2.yml` run on main with `collect_ci_evidence.py --student 2`

TO COMPLETE / JAMES HUANG

Record a green `student-3.yml` run on main with `collect_ci_evidence.py --student 3`

TO COMPLETE / MICHAEL WHITE

Record a green `student-4.yml` run on main with `collect_ci_evidence.py --student 4`

> **TO COMPLETE / DEREK SONG**
>
> Record a green `student-5.yml` run on main with `collect_ci_evidence.py --student 5`

Student 1

`student-1.yml` loads fixture data with the acquisition pipeline, then runs the tests on the feature origin (:5200) and through the shared edge (:5100):

1. `GET /api/data-platform/v1/properties/search`: contract-valid ranked matches for an accepted address; invalid queries return structured problems.
2. `/api/data-platform/v1/sources`: create, list, read, update and delete a source definition; invalid, duplicate and non-JSON bodies are rejected and not persisted.

Student 2

> **TO COMPLETE / BURHAN NAEEM**
>
> Name F2's two endpoint functions and what each test checks (about 60 words)

Student 3

> **TO COMPLETE / JAMES HUANG**
>
> Name F3's two endpoint functions and what each test checks (about 60 words)

Student 4

> **TO COMPLETE / MICHAEL WHITE**
>
> Name F4's two endpoint functions and what each test checks (about 60 words)

Student 5

> **TO COMPLETE / DEREK SONG**
>
> Name F5's two endpoint functions and what each test checks (about 60 words)

7 Testing review

`dev.py ai review testing` collects the security report and each `ci/student-N.md`, builds a deterministic checklist and reviews it through the shared loop (`review-testing.v2`). The verdict is the stricter of the model's and the checklist's; each review appends to the validation log.

Verdict	Checklist	Engine and model	Loop run	Date	Validation log
fail	10/32 passed	AI-mode, openai / gpt-5.6-terra	`0b46ed6a`	2026-10-16	3 entries

> All three pre-commit security scans were recorded. Student 1 supplied a successful CI run with endpoint coverage and no JUnit failures or errors; students 2–5 supplied no CI evidence. Documented dependency risks remain and 85 Ruff findings await owner action. (full review)

TO COMPLETE / MATTHEW SHELTON

Re-run the testing review once all five CI records exist; the committed review predates them

8 Local deployment evidence

`uv run scripts/dev.py stack up` starts the host AI tier (AI-mode, MCP, RAG and the Multi-Agent Server) and then the Compose application. Backends reach the host services through `host.docker.internal`. `scripts/capture_local_deployment.py` checks the running system, including that each AI service listens from a host process, not a container. Each server logs to its own terminal.

Local deployment report: **PASSED**, 109 of 109 checks passed, captured 2026-10-16 at commit `1a68f0a`.

Check group	Passed	Failed
Compose containers	22	0
Host AI services	9	0
Edge, frontends and health	13	0
CRUD round trips	20	0
Data	1	0
Direct AI probes	32	0
Multi-Agent terminal run	6	0
Evidence files	6	0

Terminal	Ran as	Command	Log
Docker Compose stack	foreground	`uv run scripts/dev.py stack up; stack logs`	`terminal-1-compose.log`
AI-mode host process	managed	`uv run scripts/dev.py ai serve ai-mode`	`terminal-2-ai-mode.log`
MCP host process	managed	`uv run scripts/dev.py ai serve mcp`	`terminal-3-mcp.log`

Terminal	Ran as	Command	Log
RAG host process	managed	`uv run scripts/dev.py ai serve rag`	terminal-4-rag.log
Multi-Agent host process	managed	`uv run scripts/dev.py ai serve multi-agent`	terminal-5-multi-agent.log
System test	foreground	`uv run python scripts/capture_local_deployment.py`	terminal-6-system-test.log

TO COMPLETE / MATTHEW SHELTON

Recapture the local deployment report on main at the evidence freeze; this capture ran on a branch commit

9 Cloud deployment report and review

`cloud-deployment.yml` writes the deployment report. `deploy.sh smoke` checks the home page, every feature route, one create, read and delete case per feature, and that the AI tier is disabled. `dev.py ai review cloud` reviews this evidence; a named person then records the release decision with `ai review decide`, which refuses if the evidence changed since the review.

Item	Recorded
Deployment	**SUCCEEDED** (report)
Public URL	https://propertyscope-nsw-g20.newzealandnorth.cloudapp.azure.com
Commit / image tag	`3c25c083e09e5f170767e63655593c0e5b9f1e27` / `3c25c083e09e5f170767e63655593c0e5b9f1e27`
Workflow run	https://github.com/MattShelton04/41026ASDProject/actions/runs/39112162750
AI tier	flag off (PROPERTYSCOPE_CLOUD_AI), edge reports disabled
Human release decision	**RELEASE** by Matthew Shelton, 2026-10-18 (record)

The smoke test passed all 37 checks, including create, read and delete per feature; AI-mode answered `503 ai_disabled`.

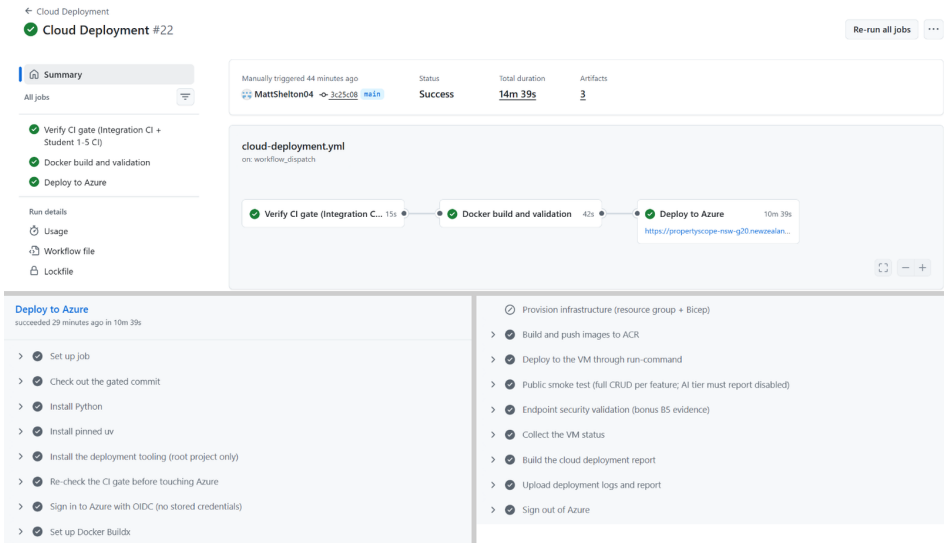


Figure 6 Gated cloud deployment run

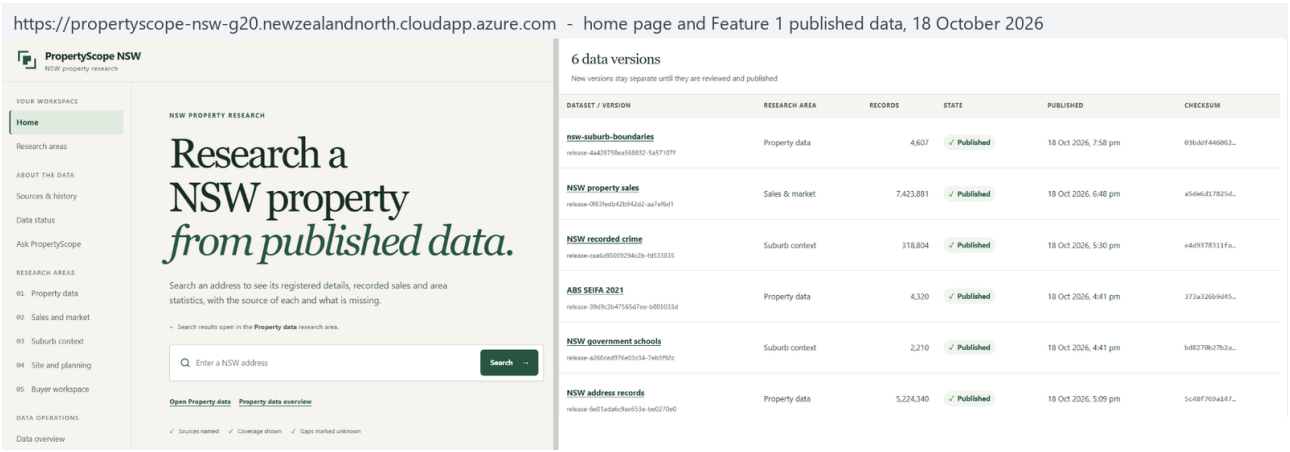


Figure 7 Deployed application with real Feature 1 data

Verdict	Checklist	Engine and model	Loop run	Date	Validation log
pass	-	deterministic, deterministic-review / checklist-review.v1	`22a7614b`	2026-10-18	2 entries

> Cloud Deployment Report Review: 18 of 18 checks passed, 0 failed (0 required), across 10 evidence file(s), 0 of them missing or unreadable. Complete: shared, student-1, student-2, student-3, student-4 and student-5. Verdict taken from the deterministic checklist without a model. (full review)

10 Multi-agent workflow evidence

The server stores runs in SQLite and appends state transitions to `workflow_history.jsonl` and handoffs, model and tool calls and decisions, with actor and time, to `coordination_audit.jsonl`. `multi-agent-server export` copies a run into the evidence folder.

Student	Run	History events	Audit events	Handoffs	Tool calls	Decisions	Final state
1	`18740ca5`	8	24	7	4	correct → partial	`partially_accepted`
1	`c17c66cc`	8	25	7	4	correct → partial	`partially_accepted`
2	Pending	-	-	-	-	-	-
3	Pending	-	-	-	-	-	-
4	Pending	-	-	-	-	-	-
5	Pending	-	-	-	-	-	-

TO COMPLETE / BURHAN NAEEM

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-2/<run> --server`

TO COMPLETE / JAMES HUANG

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-3/<run> --server`

TO COMPLETE / MICHAEL WHITE

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-4/<run> --server`

TO COMPLETE / DEREK SONG

Export a decided run with `uv run multi-agent-server export <run> --out docs/release-2/evidence/multi-agent/student-5/<run> --server`

Multi-agent workflow review

`dev.py ai review multi-agent` checks every export for all four stages, legal transitions, attributed human decisions and allowlisted tool calls.

Verdict	Checklist	Engine and model	Loop run	Date	Validation log
fail	7/35 passed	AI-mode, openai / gpt-5.6-terra	`bf3f4052`	2026-10-16	6 entries

> Student 1 has complete, correlated evidence for the required multi-agent workflow controls. Students 2–5 have no readable workflow history or coordination audit, so their Planner, Worker, Reviewer, human-decision, transition, allowlist and correlation controls cannot be verified. (full review)

Student 1

UI run `18740ca5` reviewed a real 680-record ABS CPI candidate on 10 October with the model provider. In round 1 the Reviewer recommended approve; the human chose correct, asking for completeness against the ABS source. In round 2 the Reviewer recommended correct, because the read-only tools cannot fetch ABS totals, and the human accepted step `release` only. Nothing was published. Terminal run `c17c66cc` used the same template from the CLI.

```
$ multi-agent-server export 18740ca5-5fec-482a-a279-fe179a6d25a6
01:18:30 r1              - -> planning           multi-agent-server
01:18:35 r1       planning -> working            planner
01:18:39 r1        working -> reviewing          worker
01:18:44 r1      reviewing -> awaiting_human     reviewer
01:20:46 r2 awaiting_human -> working            matthew
01:20:51 r2        working -> reviewing          worker
01:20:57 r2      reviewing -> awaiting_human     reviewer
01:21:32 r2 awaiting_human -> partially_accepted matthew
```

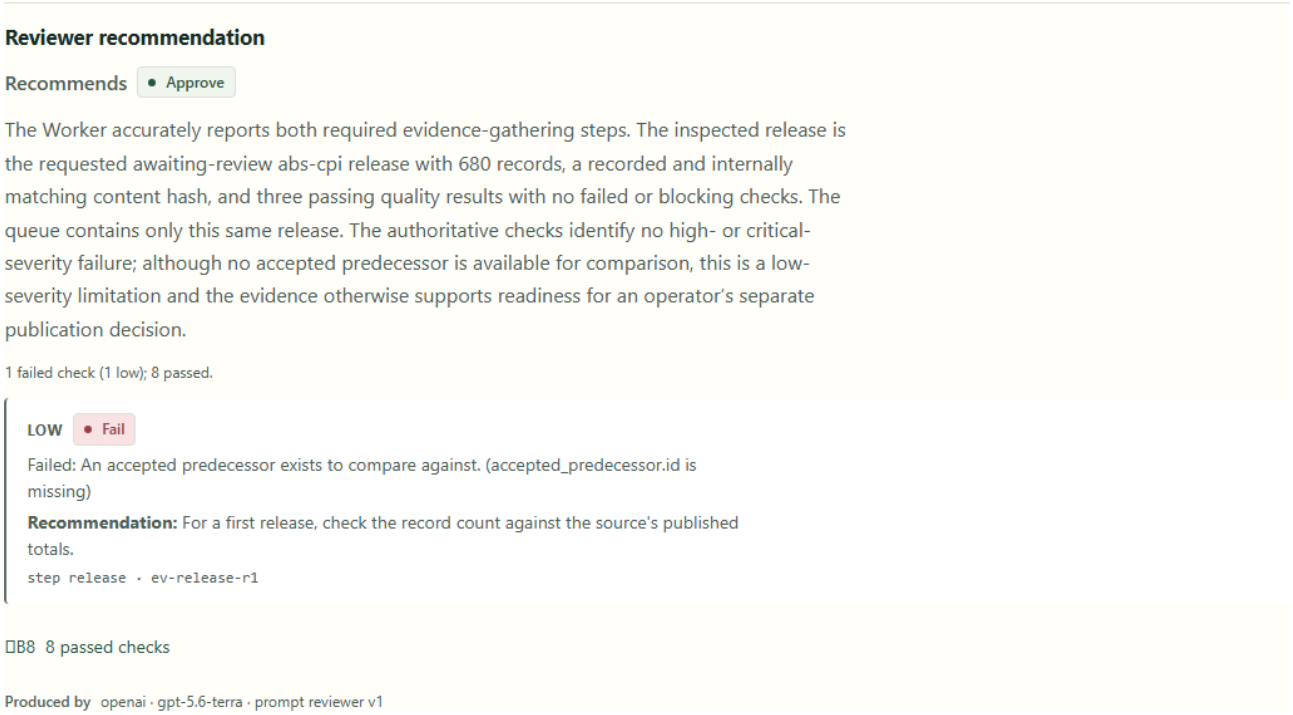


Figure 8 F1 round 1 Reviewer recommendation in the shared panel

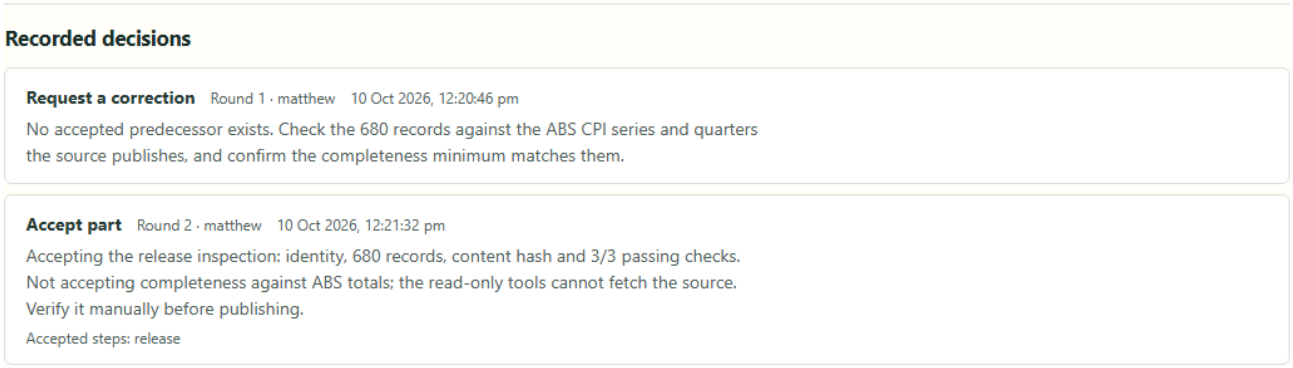


Figure 9 F1 recorded human decisions

Student 2

TO COMPLETE / BURHAN NAEEM

Summarise F2's exported run in about 50 words: input, recommendation and human decision

Student 3

TO COMPLETE / JAMES HUANG

Summarise F3's exported run in about 50 words: input, recommendation and human decision

Student 4

TO COMPLETE / MICHAEL WHITE

Summarise F4's exported run in about 50 words: input, recommendation and human decision

Student 5

TO COMPLETE / DEREK SONG

Summarise F5's exported run in about 50 words: input, recommendation and human decision

11 GitHub commit logs

Commit history: https://github.com/MattShelton04/41026ASDProject/commits/main. Section 12 lists each student's commits.

12 Contribution logs

Student 1 Matthew Shelton

Date	Commit	Release 2 work
8 Oct	36a4c26	Requirements and implementation plan
9 Oct	f86f787	Multi-Agent Server, review modes, security scans, test kit
10 Oct	bef57da, 06e0ed0	Multi-agent panel, F1 readiness review
15–16 Oct	d449a43, 07b1c81, 72f3272	CI and local evidence, report builder, quality gate
17 Oct	fdc5111, 93d0f53	Release gate, owner-attributed reviews, runbook, evidence refresh
17 Oct	5e90231, ad2b443, 71764a5, cd8e56d	Azure switch-on, site login, security audit fixes
18 Oct	b638c86, 53a2143, 65de8f6	F1 suburb-map API, unattended acquisitions, no demonstration data on Azure
18 Oct	b71c49c, 3c25c08, c1dab31	Real sales data on Azure, Cloud AI Tier workflow
18 Oct	PR #157	AI-off cloud evidence, release decision

TO COMPLETE / MATTHEW SHELTON

Replace PR #157 with its merged commit

Student 2 Burhan Naeem

TO COMPLETE / BURHAN NAEEM

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 3 James Huang

TO COMPLETE / JAMES HUANG

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 4 Michael White

TO COMPLETE / MICHAEL WHITE

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

Student 5 Derek Song

TO COMPLETE / DEREK SONG

Add a Date / Commit / Release 2 work table of your identifiable commits (about 60 words)

13 Bonus evidence

No bonus is claimed yet. Against run 39112162750, endpoint security (B5) and data security (B6) validation passed with no failures. B1–B4 have not yet run on Azure.

TO COMPLETE / MATTHEW SHELTON

For each bonus claimed, link its validation output under evidence/bonus/; otherwise keep "No bonus is claimed"

14 Showcase video

TO COMPLETE / GROUP

Link the published showcase video (at most 10 minutes) and record which students participated